

B. Math. III – Semestral Examination

Introduction to Differential Geometry

December 5, 2008

Instructions: All questions carry equal marks. The ‘dot’ on top of a function denotes its derivative.

1. Let $\gamma(t)$ be a regular curve in $\mathbb{R}^3$. Prove that its curvature κ is given by the formula:

$$\kappa = \frac{\|\ddot{\gamma} \times \dot{\gamma}\|}{\|\dot{\gamma}\|^3}$$

2. (Wirtinger’s Inequality) Let $F : [0, \pi] \rightarrow \mathbb{R}$ be a smooth function with $F(0) = F(\pi) = 0$. Prove that

$$\int_0^\pi \dot{F}^2 dt \geq \int_0^\pi F^2 dt$$

and equality holds if and only if there exists a real number A such that $F(t) = A \sin(t)$ for all $t \in [0, \pi]$.

3. Prove that the sphere of radius $r > 0$ around a point $v \in \mathbb{R}^3$ is a regular surface.

4. Define normal curvature of a curve lying on a regular surface S in $\mathbb{R}^3$. Prove that all curves lying on S and having the same tangent line at a point of S have the same normal curvature at that point.

5. Define Christoffel symbols of a parametric patch of a regular surface S in $\mathbb{R}^3$. Prove that they are invariant under local isometries.

6. Define triangulation of a regular region of a surface in $\mathbb{R}^3$. Give a triangulation of the unit sphere and of the torus and compute the Euler characteristic of the given triangulations.